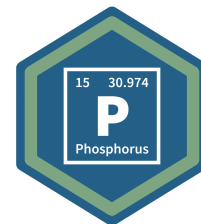


Periodic Table

(Scheduled Event - 30 Minutes)



Description: Participants will be assessed on their understanding of the periodic table, including structure, element classification, atomic structure, and periodic trends.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring: Pen/Pencil

Teams may not bring:

- Notes or guides of any kind.

Optional - teams may bring: Four-function calculator

Safety Requirements: None

The Competition: This event will include a written test and may include station-based or hands-on activities involving element identification, periodic table analysis, and data interpretation tasks.

Content topics will include but are not limited to:

1. Periodic Table Structure

- Describe organization by increasing atomic number.
- Identify periods, groups, and major regions (metals, nonmetals, metalloids).
- Identify major families: alkali metals, alkaline earth metals, halogens, noble gases, and transition metals.

2. Atomic Structure and Element Identification

- Define atomic number, atomic mass, protons, neutrons, and electrons.
- Determine subatomic particle counts for neutral atoms.
- Identify elements using atomic number, symbol, and name.

3. Periodic Trends

- Compare trends in atomic radius, electronegativity, ionization energy, and reactivity.
- Predict trends across periods and down groups.
- Use trends to infer element properties.

4. Chemical Behavior

- Explain why elements in the same group have similar properties.
- Relate valence electrons to chemical behavior and reactivity.

5. Data Analysis and Applications

- Interpret periodic tables, charts, and element property data.
- Apply periodic trends to solve problems and classify elements.

Examples:

- Identify elements using **atomic number, chemical symbol, and element name**.
- Use the periodic table to determine the number of **protons, neutrons, and electrons** in neutral atoms.
- Interpret periodic table organization by:

- atomic number
- periods (rows)
- groups/families (columns)
- metals, nonmetals, and metalloids
- Classify elements as metals, nonmetals, or metalloids based on their location and properties.
- Identify major element families:
 - alkali metals
 - alkaline earth metals
 - transition metals
 - halogens
 - noble gases
- Compare elements within the same group to predict similar properties.
- Use periodic table patterns to predict:
 - atomic size
 - reactivity
 - electronegativity
 - ionization energy
- Analyze charts or data tables comparing element properties to identify trends.
- Determine which element is more reactive based on its location in the periodic table.
- Predict whether an element is likely to form a positive ion, negative ion, or remain stable based on valence electrons.
- Use valence electron information to explain why elements in the same group behave similarly.
- Compare elements across a period to determine changes in properties.
- Interpret element property data (melting point, conductivity, reactivity, atomic size) to classify an unknown element.
- Solve scenario-based problems such as:
 - selecting the best element for a specific use
 - predicting which elements will react
 - identifying an unknown element based on properties

Scoring:

- Points will be awarded for each correct response.

Tiebreakers:

- Ties will be broken by the accuracy or quality of answers to select questions chosen by the event leader prior to competition.

Resources:

- [Periodic Table](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles